

Modeling Lens: Active Inference Without Mechanism Claims

Modeling Lens: Active Inference Without Mechanism Claims I

Active inference gives DigiPPPiP a disciplined modeling vocabulary for dyadic mark-making. Each partner maintains expectations about the shared canvas, the other partner's contribution, and the evolving meaning of the artifact. The shared drawing is therefore a sensory sample, a prediction target, and an action channel [friston2010fep;friston2017process;ramstead2020two].

Active-inference accounts of human communication support this interactional reading only when the model stays tied to observable communicative events rather than becoming a generic metaphor [vasil2020communication]. In second-person terms, the partner is not a stimulus to observe from outside but an agent whose responses shape one's own policy choices [schilbach2013secondperson;redcay2019secondperson;bolis2024secondperson].

Modeling Lens: Active Inference Without Mechanism Claims II

The main claim is intentionally narrow. DigiPPPiP does not claim that participants literally compute free-energy equations, nor that the current simulations measure a real couple. The claim is that a useful formal account must specify states, observations, policies, update rules, and observable consequences. If a study cannot map interface events to observations, define partner-relevant policies, or distinguish reciprocal updating from shared-stimulus effects, then active inference is not doing explanatory work for that study.

Modeling Lens: Active Inference Without Mechanism Claims III

The formal appendix consolidates the mathematical primitives, model figures, and toy diagnostics in one place [[@sec:formalisms_appendix](#)]. That appendix maps DigiPPPiP events to active-inference variables, gives the Gaussian point-belief model used by `src/active_inference.py`, shows the coupled-versus-decoupled toy run, and places narrative entropy, epistemic arcs, and geometric hyperscanning under the same evidence boundary. Keeping those details out of the main line makes the argument easier to inspect: the manuscript first states why the model is useful, then the appendix shows exactly what the model assumes.

Modeling Lens: Active Inference Without Mechanism Claims IV

This separation also protects the empirical agenda. Hyperscanning, narrative information, and free-energy trajectories are candidate measurement languages, not outcome evidence. Synchrony can be inflated by shared stimulus timing, motion artifacts, physiology, preprocessing decisions, and analytic windows [tachtsidis2016fnirs;hamilton2021hyperscanning;zimmermann2024methodological]. Narrative entropy can identify symbolic structure without proving shared meaning. A conceptual free-energy curve can expose a coupling assumption without showing that partners improved. The framework is strongest when each formalism names the future comparison that could reject it.

Modeling Lens: Active Inference Without Mechanism Claims V

The practical upshot is a modeling checklist. A DigiPPPiP study should log the visible mark, the timing of the response, the partner-facing cue, the repair or withdrawal option, and the artifact-governance state. It should then ask whether the model explains something beyond simpler alternatives such as ordinary shared drawing, video conversation, or solo art-making. Active inference remains in the manuscript because it forces those commitments into the open, not because it upgrades conceptual simulations into evidence.